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HALL EFFECT AND THE BEAUTY AND CHALLENGES OF SCIENCE

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I. INTRODUCTION

It is a great honor to be asked to participate in this commemorative symposium on the centennial of the discovery of the Hall effect here at Johns Hopkins. The Hall effect is a superb example of a phenomenon which, while simple in initial concept, is really very profound in its implications and application to a variety of solids of inherently differing characteristics. The evolution of the versatility of Hall data in interpreting complex transport in solids is indeed a classic example of the beauty and challenges of science. Let us try to follow some of the major developments in this progression of theoretical and experimental achievements.

II. LORENTZ FORCE

Fundamentally the Hall effect is a manifestation of the Lorentz force on an ensemble of charged particles constrained to move in a given direction and subjected to a transverse magnetic field. The physics involved is illustrated in Figure 1 by consideration of a confined stream of free particles, each having a charge e and a steady-state velocity v_x due to the impressed electric field E_x. A magnetic field in the z direction produces a force F_y in the y direction in accordance with Eq.(1), below, which gives the force on a charged particle subjected to an electric field E and a magnetic field H:

$$\vec{F} = e\,[\vec{E} + (1/c)\vec{v} \times \vec{H}] \text{ (Gaussian units).} \qquad (1)$$

However, the boundary condition imposed by the orientation of the

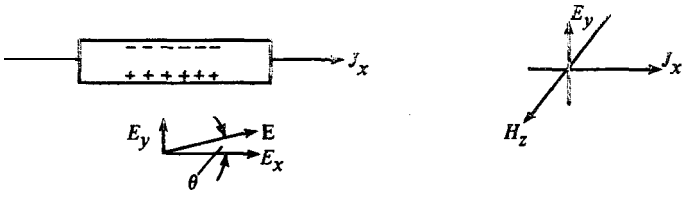


Fig. 1. Illustration of Hall field E_y due to action of magnetic field on a confined streamof positive charge carriers. For the example shown, e, J_x, and H_z are positive--as are also v_x E_x, and E_y . The resultant electric field E and the Hall angle θ are shown at the lower left. In the case of electrons, the Hall field is reversed for positive J_x, both v_x and e then being negative.

specimen requires that the transverse current J_y vanish, and hence at steady state F_Y will vanish. Therefore atransverse electric field E_y must build up in order to counteract the Lorentz force, as is indicated in Eq.(2):

$$F_y = 0 = E_Y - (1/c)v_x H_z. \qquad (2)$$

This transverse field is known as the Hall field, and we shall designate it by E^H. Its magnitude is given by Eq.(3), where we have also expressed the charge carrier velocity in terms of current density J_x and charge carrier concentration n:

$$E^H(\equiv E_y) = (1/c)v_x H_z = J_x H_z/nec, \quad \text{using } J_x = nev_x. \qquad (3)$$

The Hall field can be related to the current density by means of the Hall coefficient R, as noted below:

$$E^H = RJ_x H_z, \qquad R = 1/nec \quad \text{(Gaussian units*)}, \qquad (4)$$

*In the Gaussian system, mechanical quantities are expressed in cgs units, electrical quantities in esu, and magnetic fields in gauss or oersteds. In the practical system, mechanical quantities are in cgs units, electrical quantities in volts and coulombs, and magnetic fields in gauss or oersteds. In the equations involving transport phenomena, conversion from the Gaussian system to the practical system is effected by replacing c by unity and replacing H by $H/10^8$, where those quantities explicitly occur. The practical units for R are $cm^3/coulomb$; those for mobility are $cm^2/volt\text{-}sec$.

$$E^H = RJ_x H_z/10^8, \qquad R = 1/ne \text{ (practical units*)} \qquad (4a)$$

Our sign convention is that e < 0 for electrons.

Thus we see that for our simple model, the Hall coefficient is inversely proportional to the charge carrier density n, and that it is negative for electron conduction and positive for hole conduction. Finally, referring again to Fig. 1, we can define a Hall angle by the relation

$$\theta = \tan^{-1}(E_y/E_x). \qquad (5)$$

An important point to be made here is that the Hall effect may be regarded as the rotation of the electric field vector in the sample as a result of the applied magnetic field. More will be said later of the Hall angle.

In view of the relations we have presented thus far, some of the readers will perhaps be tempted to utter the cliche "So what could be simpler?" To which I shall respond that, as this talk progresses, it will be seen to be more appropriate to borrow instead from Shakespeare his classic phrase "How infinite in faculties!"

III. PRACTICAL CONSIDERATIONS

In perfecting the use of Hall phenomena as a tool for studying the electronic characteristics of solids, one can identify two classes of complications The first has to do with the experimental determination of the Hall coefficient, and the second with its interpretation in terms of the electronic character of the solid.

1. Measurement Techniques

With regard to measurement of the Hall coefficient, a number of techniques are available.[1-6] The conventional procedure is to measure, by means of a potentiometer or other high-impedance device, the Hall voltage V^H across a parallelepiped in a direction normal to both $\vec{H}$ and $\vec{J}$. The arrangement is shown in Figure 2. The expression for the Hall coefficient, given in Eq.(6), follows directly from the previous equation (4a) when the geometry of the sample is taken into account:

$$R = 10^8 V_{3,4}\, t/IH \qquad \text{(practical units)}. \qquad (6)$$

* See footnote on previous page.

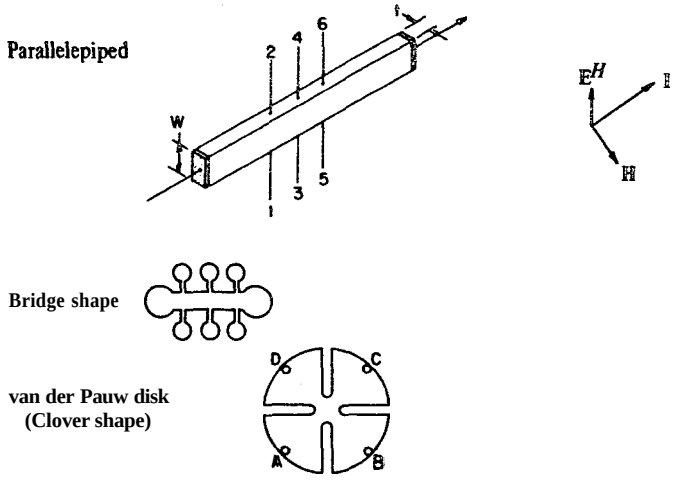


Fig. 2. Measurement samples for Hall coefficient and related effects.
Top: Conventional parallelepiped. Hall probes are 3 and 4; resistivity probes are 2 and 6 and also 1 and 5.
Middle: Typical form of "bridge-type" sample.
Bottom: Clover-shaped design, or van der Pauw disk.

Here t is the thickness of the sample, measured along $\vec{H}$, $\vec{I}$ is the current (as contrasted to the current density, which we call **J**), and $V_{3,4}$ is the Hall voltage V^H. Inasmuch as V^H may be the order of microvolts, extreme care must be taken to avoid extraneous voltages. An example is the misalignment voltage, caused by probes 3 and 4 not being on an equipotential plane when H = 0. A somewhat common procedure for eliminating this spurious voltage is to do measurements for opposite directions of H and take half the algebraic difference, inasmuch as the Hall voltage is odd in H. The misalignment voltage is, of course, independent of H but proportional to I. As we shall see later, however, if anisotropic solids are being studied and various galvanomagnetic effects are to be measured, then it is desirable to avoid any misalignment voltage by adjusting the position of the Hall probes to obtain a null reading when H = 0. An alternative technique involves replacing probe 4 by a sliding contact [1,5] to a resistor which is attached to the sample at two points, such as 2 and 6 in Figure 2. Disposing of misalignment voltages in this fashion allows one to use the reversal of magnetic field to separate even and odd galvanomagnetic effects.

Also, reversals of H and I can be used to eliminate unwanted voltages resulting from certain thermal effects. Such spurious voltages can arise from temperature gradients and resulting thermoelectric or thermomagnetic effects, as will be discussed later.

Connections to the sample shown in the sketch should be arranged so as to avoid, as much as possible, perturbations of the electric field in the sample. Thus the potential probes are small points. The end contacts, which short the Hall field, should be sufficiently far removed from the Hall potential probes. The latter condition can be ensured for sample length-to-width ratios of at least four.[1] Hall samples can also be cut in a bridge-shaped design, as is shown in the figure. This has the advantage of providing large contact areas with a reasonably small disturbance of the electric field in the sample. Another popular design is the van der Pauw disc.[9] A number of other arrangements are presented in the literature,[10] including methods for determining the Hall coefficient without contacting the specimen.[10a] Information on profiling techniques is also available.[10b]

2. General Transport Phenomena

The Hall effect is but one manifestation of a variety of electrical transport phenomena occurring in solids in a magnetic field, which are referred to as galvanomagnetic effects and thermomagnetic effects. To illustrate these effects, we write the phenomenological equations which relate the densities of electric and thermal currents to the gradients producing the transport (that is, electric fields and temperature gradients).[11]

$$J_i = \sum_j \sigma_{ij}(\vec{H}) E_j^* + \sum_j M_{ij}(\vec{H}) \frac{\partial T}{\partial x_j}, \qquad (7)$$

$$j = 1,2,3$$

$$q_i = \sum_j N_{ij}(\vec{H}) E_j^* + \sum_j L_{ij}(\vec{H}) \frac{\partial T}{\partial x_j}. \qquad (8)$$

Summations are of course carried out over the three coordinate directions, and the four coefficients in the above equations are tensor quantities. which are magnetic-field dependent. The σ_{ij} are the components of the electrical conductivity tensor, and the other coefficients are related to thermoelectric phenomena and to thermal conductivity. The field $\vec{E}^*$ is the negative gradient of the electrochemical potential, namely

$$E_j^* = -\frac{1}{e}\frac{\partial(e\phi + \zeta)}{\partial x_j} = E_j - \frac{1}{e}\frac{\partial \zeta}{\partial x_j}, \qquad e < 0 \text{ for electrons.} \quad (9)$$

In the case where there are no temperature or carrier concentration

variations throughout the sample, then the chemical potential ζ is constant and $\vec{E}^*$ reduces to the electric field E, that is, the negative gradient of the electrostatic potential ϕ.

Since most measurements are carried out under conditions where $\vec{J}$ is the independent variable, it is desirable to invert[12] the electrical current density equation, which process yields relations (10) and (11) below

$$E_i^* = \sum_k \rho_{ik}(\vec{H}) J_k + \sum_k \alpha_{ik}(\vec{H}) \frac{\partial T}{\partial x_k}, \qquad k = 1,2,3 \tag{10}$$

$$q_i = \sum_k \pi_{ik}(\vec{H}) J_k - \sum_k \kappa_{ik}(\vec{H}) \frac{\partial T}{\partial x_k}. \tag{11}$$

Here the respective tensors represent well-known coefficients, namely: p, resistivity; a, thermoelectric power (Seebeck coefficient); π, Peltier coefficient; and κ, thermal conductivity. For the sample orientation such that $J \equiv J_x$, a number of the terms in the summations drop out; and with further simplifying assumptions including $H \equiv H_z$, the preceding equations reduce to

$$E_y^* = \rho_{yx}(H) J_x + \alpha_{yx}(H) \frac{\partial T}{\partial x} + \alpha_{yy}(H) \frac{\partial T}{\partial y}, \tag{11a}$$

$$E_x^* = \rho_{xx}(H) J_x + a_{xx}(H) \frac{\partial T}{\partial x} + \alpha_{xy}(H) \frac{\partial T}{\partial y}, \tag{11b}$$

$$q_x = \pi_{xx}(H) J_x - \kappa_{xx}(H) \frac{\partial T}{\partial x} - \kappa_{xy}(H) \frac{\partial T}{\partial y}, \tag{11c}$$

$$q_y = \pi_{yx}(H) J_x - \kappa_{yx}(H) \frac{\partial T}{\partial x} - \kappa_{yy}(H) \frac{\partial T}{\partial y}, \tag{11d}$$

$$J \equiv J_x, \quad H \equiv H_z, \quad \rho_{xz} = \rho_{yz} = \rho_{zx} = \rho_{zy} = 0.$$

With the current restricted to the x direction and the magnetic field lying along the z direction, the "further simplifying assumptions" mentioned above include the premise that all tensor components involving index z vanish. This condition is satisfied, for example, in isotropic systems or in cubic systems with coordinate axes along the cube axes. As a result of these constraints, there are no gradients in the z direction. In anisotropic media in general, however, the limitation of the gradients to two dimensions may not be possible.[13]

At this point we want to emphasize that there exists a set of general but very powerful relations formulated by Kohler and Onsager, which apply to the resistivity and thermal conductivity

tensors and which connect the Peltier and Seebeck coefficients,[14] namely:

$$\rho_{ij}(\vec{H}) = \rho_{ji}(-\vec{H}), \qquad \kappa_{ij}(\vec{H}) = \kappa_{ji}(-\vec{H}), \tag{12}$$

$$\pi_{ij}(\vec{H}) = T\alpha_{ji}(-\vec{H}). \tag{13}$$

As we shall examine in more detail later, these Kohler-Onsager reciprocal relations tell us that the symmetric parts of the ρ_{ij} and the κ_{ij} tensors [Eq.(12)] must be even in H and the antisymmetric parts odd in H.

Going back now to Eqs.(lla) to (11d) we see that in general the electric fields and temperature gradients are connected via the thermoelectric coefficients. This is an example of how life can be complicated. For by Eq.(lla) we see that a measured Hall field will in general contain contributions arising from temperature gradients and entering through the thermoelectric and thermomagnetic coefficients. Inasmuch as the usual theoretical relationships involving the Hall effect are based on isothermal conditions, we now have two choices: (a) d0 measurements in a manner so as to achieve reasonable approaches to isothermal conditions, or to correct for or to cancel or balance out the thermoelectric and thermomagnetic voltages. For certain cases, cancellation can be achieved through appropriate reversals of current and magnetic field: The other alternative is (b) to enlarge the theory and take account of the complications introduced through Eqs.(lla) to (lld). This approach would only be feasible for simple isotropic solids. As we shall see presently, the theory even for isothermal Hall effect can be very complicated, except for ideal materials. Nevertheless, an adiabatic Hall coefficient is sometimes measured. In this case, the boundary condition $\partial T/\partial y = 0$ is replaced by $q_y = 0.15$, 15a

To shed some light on what particular properties of a solid make for large differences between isothermal and adiabatic measurements, let us consider the simple example of the resistivity at zero magnetic field. Eqs.(llb) and (11c) then yield, for the isothermal and adiabatic measurement conditions, the following relationships:

$$\text{Isothermal:} \quad \begin{cases} E_x^* = \rho J_x \\ q_x = \pi J_x \end{cases}, \quad \text{grad } T = 0 \tag{14}$$

$$E_x^* = \rho J_x + \alpha \partial T/\partial x$$

Adiabatic: (15)

$$0 = q_x = \pi J_x - \kappa \partial T/\partial x.$$

Therefore, for a determination of resistivity under the isothermal and adiabatic conditions, the following expressions apply:

$$\rho^i \equiv (E_x^*/J_x)_{T=\text{const}} = \rho, \quad (16)$$

$$\rho^{ad} \equiv (E_x^*/J_x)_{qx=o} = \rho(1 + \pi\alpha/\kappa\rho) \equiv \rho(1 + Z^*), \quad (17)$$

where

$$Z^* \equiv \pi\alpha/\kappa\rho = \alpha^2 T/\kappa\rho \quad \text{(thermoelectric figure of merit)}. \quad (18)$$

Hence,

$$(\rho^{ad}-\rho)/\rho = Z^*. \quad (19)$$

The thermoelectric figure of merit Z*, defined in Eq.(18), is a measure of the efficiency of the material in a thermoelectric cooling or power generation process. We thus note that it is the thermoelectric figure of merit which is an important consideration in the difference between data taken under isothermal and adiabatic conditions. It is to be pointed out that the above analysis was done for one-dimensional transport, namely resistivity. The situation for Hall effect would be more complex. Nevertheless, we can expect differences between isothermal and adiabatic Hall effects to be small for materials where Z* << 1. For example, in materials with the diamond or zincblende crystal structure, where the lattice contribution to the thermal conductivity is relatively large, the correction to isothermal coefficients for specimens measured under adiabatic conditions is quite small-often under one percent.[15]

Going back again to option (a), we wish to point out that one can reduce temperature gradients along the sample by the use of large end-contacting blocks of copper or other material with a large thermal conductivity, by use of isothermal baths, or-in the more difficult cases-by resorting to a-c measurements.[16] An effective technique in the latter case is to use square wave currents of alternating polarity.[17] In d-c measurements, the appropriate switching in polarity of both magnetic field and sample current is useful in some instances.

Before leaving this topic, I should like to point out how thermomagnetic effects can arise on the basis of our simple free-electron model. The Lorentz force acts in principle as a velocity selector-the slow electrons being deflected less than the more energetic ones-a process which of course creates a temperature gradient in the transverse direction. This temperature difference will result in a transverse potential difference due to the Seebeck coefficient of the material. The former phenomenon is called the Ettingshausen effect. Detailed information on the various thermoelectric and thermomagnetic phenomena is available in numerous places.[17a]

3. Galvanomagnetic Effects, Anisotropic Solids

We now concentrate on isothermal conditions and discuss in more detail the effects of a magnetic field on electrical transport. In particular, we consider the major galvanomagnetic effects in the general case of anisotropic solids. For a specimen with essentially no temperature gradients nor composition variations, the distinction between the gradient of the electrochemical potential, $\vec{E}^*$, and the electric field $\vec{E}$ vanishes. Equations (11a) to (11d) now reduce to the "short form" shown below:

$$E_i = \sum_k \rho_{ik}(\vec{H}) J_k. \quad (20)$$

The words "short form" will probably cause some eyebrow raising after one looks at Eq.(21), which represents the expansion of the resistivity tensor in powers of the magnetic field.

$$\rho_{ik}(\vec{H}) = \rho^o_{ik} + \sum_\ell \rho^o_{ik\ell} H_\ell + \sum_{\ell,m} \rho^o_{ik\ell m} H_\ell H_m + \cdots \quad (21)$$

The superscript zero on a tensor component indicates that it is independent of magnetic field. Even a casual glance at Eq.(21) reveals that. in spite of the neglect of terms higher than second order in H, the resistivity tensor for an anisotropic material can be a complex creature-especially if the magnetic field is not along a direction of high symmetry that can cause the coefficients of an appreciable number of terms in each summation to vanish.

We shall identify the linear terms in H in Eq.(21) with the Hall effect, those quadratic in H with magnetoresistivity, and further terms as higher order effects of these phenomena. However, such a convention does not appear universally throughout the literature. Some of the quadratic terms-i.e. those ones responsible for an electric field normal to the current-have been identified with a quadratic "Hall effect"[18] and also a planar "Hall effect".[19] The latter phenomenon exists even in isotropic solids-as we shall see later. To avoid these problems in definition, and to establish a base to see how the complications come about, we shall adopt what seems to me to be the most straightforward of the various

conventions. This is to divide the resistivity tensor into antisymmetric and symmetric parts and to identify the antisymmetric part with the Hall effect, and the symmetric part with generalized magnetoresistivities. This division is written as follows:

$$\rho_{ik}(\vec{H}) = \rho^a_{ik}(\vec{H}) + \rho^s_{ik}(\vec{H}), \tag{22}$$

where

$$\rho^a_{ik}(\vec{H}) = -\rho^a_{ki}(\vec{H}); \qquad \rho^s_{ik}(\vec{H}) = \rho^s_{ki}(\vec{H}). \tag{23}$$

The components of the general Hall field E^a are then given by

$$E^a_i(\vec{H}) = \sum_k \rho^a_{ik}(\vec{H}) J_k. \tag{24}$$

Now an expression involving an antisymmetric tensor such as Eq.(24) can be written in terms of a vector cross product, as was done by Casimer[20] and is shown in Eq.(25):

$$\vec{E}^a(\vec{H}) = \vec{R}(\vec{H}) \times \vec{J}. \tag{25}$$

The quantity $\vec{R}(\vec{H})$ is called the Hall vector. We note, in view of the Kohler-Onsager relations, [Eq.(12)], that $\vec{R}(\vec{H})$ is odd in $\vec{H}$. It likewise follows that the $\rho^s_{ik}(\vec{H})$, which we identify with magnetoresistivity, are even in $\vec{H}$.

$$\vec{R}(\vec{H}) = -\vec{R}(-\vec{H}), \qquad \rho^s_{ik}(\vec{H}) = \rho^s_{ik}(-\vec{H}). \tag{26}$$

The components of the Hall vector can be expressed in terms of the antisymmetric components of the resistivity tensor as follows:[20a]

$$R_k(\vec{H}) = \frac{1}{2}\sum_{\ell,m} \epsilon_{k\ell m}\, \rho^a_{m\ell}(H). \tag{27}$$

The $\epsilon_{k\ell m}$ are the permutation tensors, which are ± 1 when $k\ell m$ are all different and zero otherwise. They are positive when the indices are in standard cyclic order, negative for the reverse case. The factor 1/2 occurs because the tensor $\rho^a_{m\ell}$ is summed over its antisymmetric indices.

Looking again at Eq.(25), we make the startling discovery that while the Hall field is always perpendicular to $\vec{J}$, it is not necessarily perpendicular to $\vec{H}$. We thus need to distinguish between the conventional Hall field (i.e. the transverse Hall field), which is perpendicular to both $\vec{J}$ and $\vec{H}$, and another component of $\vec{E}^a(\vec{H})$ which is perpendicular to both $\vec{J}$ and the transverse Hall field.

The phenomenon in this latter case is called the longitudinal Hall effect. The two contributions to the general Hall field, and their components along unit vectors in the appropriate directions, are shown below:

$$\vec{E}^a(\vec{H}) = \vec{E}^H(\vec{H}) + \vec{E}^{H,L}(\vec{H}), \tag{28}$$

$$= E^H(\vec{H})\frac{\vec{H}\times\vec{J}}{|\vec{H}\times\vec{J}|} + E^{H,L}(\vec{H})\frac{\vec{J}\times(\vec{H}\times\vec{J})}{|\vec{J}\times(\vec{H}\times\vec{J})|}. \tag{29}$$

Explicit expressions for the conventional and longitudinal Hall fields can be put in the following useful form:

$$\text{Transverse:}\quad \vec{E}^H = \frac{\vec{E}^a(\vec{H})\cdot\vec{H}\times\vec{J}}{|\vec{H}\times\vec{J}|}\,\frac{\vec{H}\times\vec{J}}{|\vec{H}\times\vec{J}|}, \tag{30}$$

$$\text{Longitudinal:}\quad \vec{E}^{H,L} = \frac{\vec{E}^a(\vec{H})\cdot\vec{J}\times(\vec{H}\times\vec{J})}{|\vec{J}\times(\vec{H}\times\vec{J})|}\,\frac{\vec{J}\times(\vec{H}\times\vec{J})}{|\vec{J}\times(\vec{H}\times\vec{J})|}. \tag{31}$$

In view of the identity

$$\vec{J}\times(\vec{H}\times\vec{J}) \equiv J^2\vec{H} - \vec{J}\cdot\vec{H}\,\vec{J}, \tag{32}$$

we see that if $\vec{H}$ is perpendicular to $\vec{J}$, then the longitudinal Hall field will lie along $\vec{H}$.

It is of obvious interest to determine the principal components of the Hall vector. To do this, we use Eqs.(25) and (29) to obtain

$$\vec{R}(\vec{H})\times\vec{J} = \frac{E^H(\vec{H})}{|\vec{H}\times\vec{J}|}\vec{H}\times\vec{J} + \frac{E^{H,L}(\vec{H})}{|\vec{J}\times(\vec{H}\times\vec{J})|}(\vec{J}\times\vec{H})\times\vec{J}. \tag{33}$$

From which it follows at once that

$$\vec{R}(\vec{H}) = \frac{E^H(\vec{H})}{|\vec{H}\times\vec{J}|}\vec{H} + \frac{E^{H,L}}{|\vec{J}\times(\vec{H}\times\vec{J})}\vec{J}\times\vec{H}. \tag{34}$$

Thus we see that for the conventional Hall effect, the Hall vector lies along $\vec{H}$; for the longitudinal Hall effect, it lies along $\vec{J}\times\vec{H}$. We are now in a position to express the Hall vector, $\vec{R}^T(\vec{H})$, for the conventional (transverse) Hall effect in terms of the usual Hall coefficient $R(\vec{H})$. We have, from Eq.(34),

$$\vec{R}^T(\vec{H}) = \frac{E^H(\vec{H})}{|\vec{H}\times\vec{J}|}\vec{H} = R(\vec{H})\vec{H}, \text{ where } R(\vec{H}) \equiv \frac{E^H(\vec{H})}{|\vec{H}\times\vec{J}|}. \tag{35}$$

Thus the familiar Hall field becomes

$$\vec{E}^H(\vec{H}) = R(\vec{H})\vec{H}\times\vec{J}. \tag{36}$$

Of course we knew relation (36) all along-way back from Eq.(4)— but it's gratifying to have it readily spring forth from our general and somewhat esoteric derivation involving the antisymmetric components of the resistivity tensor. The longitudinal Hall effect has been studied theoretically by Grabner,[21] with experimental data for n-type germanium. He finds that it vanishes for (a) spherical energy surfaces. (b) when $\vec{H}$ is parallel to an axis of rotation of the crystal, and (c) in the limit of strong magnetic fields.

In order to examine in more detail the various galvanomagnetic phenomena which produce electric fields in directions coincident with the Hall field, we examine the expansion of the components of the resistivity tensor $\rho(\vec{H})$, as was done in Eq.(21). The expression for the electric field, through second order terms in H, is

$$E_i = \sum_k \rho^o_{ik} J_k + \sum_{k,\ell} \rho^o_{ik\ell} J_k H_\ell + \sum_{k,\ell,m} \rho^o_{ik\ell m} J_k H_\ell H_m + \cdots \qquad (37)$$

Here I repeat again our convention that we associate the terms that are odd in H with the Hall effect and those that are even with magnetoresistive fields To simplify matters, we now place the current density vector J along axis 1 and the magnetic field H along axis 3. so that the expression for the conventional Hall field reduces to (38) below, and the longitudinal Hall field is given by Eq.(39).

$$E^H = E_2 = \rho^o_{213} H_3 J_1 , \qquad (38)$$

$$H \equiv H_3, \qquad J = J_1$$

$$E^{H,L} \equiv E_3 = \rho^o_{313} H_3 J_1 . \qquad (39)$$

Cubic crystal with cube axes along 1,2,3

$$\rho^o_{ik\ell} = \epsilon_{ik\ell} \rho^o_{123} . \qquad (40)$$

For a material with cubic symmetry, if $\vec{H}$ is along a cube axis then there is only one value for the Hall coefficient, as is indicate2 by Eq.(40). The longitudinal Hall field vanishes. However, if H is not along a cube axis, then as Grabner showed, ρ^o_{313} will not vanish so long as the energy surfaces are nonspherical .

To see how additions to the Hall fields can arise via the magnetoresistive coefficients, we consider the contributions to E_2 and E_3 from the second order terms in H of Eq.(37). We have

$$E_2 = \rho^o_{2133} H_3^2 J_1 , \qquad (41)$$

$$H \equiv H_3, \qquad J \equiv J_1$$

$$E_3 = \rho^o_{3133} H_3^2 J_1 . \qquad (42)$$

The above contributions vanish if H is along a crystal axis of sufficient symmetry. For example, if H is along a cube axis in the cubic system or along the three-fold axis in a trigonal system, the tensor components ρ^o_{2133} and ρ^o_{3133} vanish. However, they are non-zero in a trigonal crystal for certain magnetic field directions that are normal to the three-fold axis. Phenomena depicted by Eq.(41) have been referred to in early literature as the quadratic "Hall effect".[18] The recent tendency is, as we have enunciated, to consider the effect as a magnetoresistive contribution inasmuch as it arises from the symmetric part of the resistivity tensor.

Let us now recapitulate a bit. We recall that by our definition the conventional Hall field is along $\vec{H} \times \vec{J}$, and is therefore perpendicular to both $\vec{J}$ and $\vec{H}$. However, it is not necessary that H be perpendicular to $\vec{J}$, just so long as it is not parallel. For simplicity, though, the experimenter usually orients the magnetic field normal to $\vec{J}$. For such case, the simplified equations (38) to (42) are applicable. There are, however. examples in the literature where the only condition imposed is that $\vec{E}^{meas}$ be normal to $\vec{J}$.[22] For such a case, magnetoresistive coefficients can contribute to $\vec{E}^{meas}$ even in isotropic solids. To see how this can come about, we again choose $\vec{J}$ along axis 1, $\vec{E}^{meas}$ along 2, hut now permit $\vec{H}$ to have an arbitrary direction, as indicated by the constraints shown as Eq.(42a) below. Equation (37) then yields the results for E^{meas} ($\equiv E_2$) shown in Eqs.(43) and (44).

$$J = J_1, \quad E^{meas} \equiv E_2, \quad H \equiv H(H_1, H_2, H_3), \qquad (42a)$$

$$E_2^{(odd)}/J = \rho^o_{211} H_1 + \rho^o_{212} H_2 + \rho^o_{213} H_3 , \qquad (43)$$

$$E_2^{(even)}/J = \rho^o_{2111} H_1^2 + \rho^o_{2122} H_2^2 + \rho^o_{2133} H_3^2 + \left| 2\overline{\rho^o_{2112}} H_1 H_2 \right| +$$

$$2\,\rho^o_{2113} H_1 H_3 + 2\,\rho^o_{2123} H_2 \qquad (44)$$

These equations, which can be applied to an arbitrary crystal orientation, show how complicated the expressions can be when the currents and magnetic fields are not along directions of sufficient symmetry to cause most of the galvanomagnetfc coefficients to vanish. Of course, as we noted before, if H is along a cube axis in, say, germanium then only the third term in Eq.(43), namely the conventional Hall term, is non-zero. But if the direction of $\vec{H}$ is arbitrary with respect to directions of crystal symmetry, we can get a longitudinal Hall field by means of the other terms.

With regard to contributions to E_2 from magnetoresistive coefficients, results are much more formidable, as is seen in Eq.(44).[22a] Again, for a cubic crystal with H along a cube axis

only the term in the box is non-zero. In a trigonal system, we can get one or two additional terms depending upon which coordinate is aligned along the three-fold axis. We saw examples of some of these phenomena in Eqs.(41-42) earlier, where we mentioned that the effects had bee" called a quadratic Hall effect" in early literature. We now concentrate on the boxed term in Eq.(44). Interestingly enough, the galvanomagnetic coefficient ρ^{o}_{2112} is non-zero even in isotropic systems. Since this coefficient describes the state of affairs when the current and both fields are confined to a single plane the phenomenon is sometimes known as the planar "Hall effect".[19,22a] It is usually expressed in terms of the angle between H and J, thus

$$E_2 = 2\rho^{o}_{2112}H_1H_2J = \rho^{o}_{2112}JH^2 \sin 2\phi, \qquad \phi \equiv \sphericalangle H,J. \tag{45}$$

Some authors have referred to the planar "Hall effect" as the "pseudo Hall effect" since it arises from a magnetoresistive coefficient (i.e. the symmetric part of the resistivity tensor), and therefore according to our definition it is not a Hall effect. However, both it and the so-called quadratic "Hall effect" produce electric fields that are normal to the current. Therefore care must be exercised not to confuse such fields with the conventional Hall field. Inasmuch as these magnetoresistive contributions are even in H, they can be eliminated by taking data for both directions of magnetic field.

4. Specimen Homogeneity

A major consideration in the measurement and interpretation of Hall data is the homogeneity of the sample. This is especially important in the case of semiconductors where the densities of impurities and crystal imperfections are kept small. Thus any deviations from such a state are especially conspicuous. In addition, most semiconductors of interest have relatively large mobilities, and a large mobility accents all galvanomagnetic effects. We shall discuss the effect of inhomogeneities in more detail in the next part of this talk. We shall see there that inhomogeneities can produce an intermixing of Hall and magnetorrsistivity effects and can thus cause very large magnetic field dependencies of the Hall coefficient. We also wish to point cut that the transverse magnetoresistance can really go wild, since this effect is very sensitive to any perturbation of the Hall field.[23] Although time does not permit me to include magnetoresistance in any detail in this talk on Hall effect, I would point out that it is eve" more sensitive than the Hall effect to scattering processes, anisotropy, band structure, inhomogeneities, etc. The trouble is that it is so sensitive to so many things that results are difficult to interpret. Nevertheless, magnetoresistance measurements are an important adjunct to Hall data when a thorough study is being made of a

semiconductor. With regard to inhomogeneities, it is obviously very important to discover their existence. Ways of doing this include checks of resistivity as a function of position in the bulk material with a four-point probe, comparison of data on measured samples cut from different locations, and analyses of magnetic-field dependence of Hall and other galvanomagnetic effects.[24a]

5. summary to Part III

It is now appropriate to summarize some of the major considerations we have discussed during the first part of this talk. We showed that, although measurements of the Hall effect are relatively straightforward, there are a number of areas where care needs to be exercised. These include voltage probe misalignment problems, end-contact shorting, unwanted contributions due to thermoelectric and thermomagnetic effects, and, in anisotropic solids, contributions from magnetoresistive effects which are collinear with the Hall field but are eve" in H. Lastly, but always to be kept in mind, especially if unfamiliar materials are being studied, are effects of sample inhomogeneities.

IV. THEORETICAL ASPECTS

For the remaining part of this paper, I want to discuss theoretical aspects involved in the interpretation of Hall data in terms of the electronic character of the solid. We must thus make the transition from the simple free-particle treatment discussed at the beginning of this talk to the state of affairs existing in a real solid. For example, we need to take account of the distribution of velocities and the interactions of the charge carriers with impurities, defects, and lattice thermal vibrations of the solid (i.e. the scattering), as well as the band structure of the material. The latter consideration relates to the fact that the charge carriers in the solid are not free but exist in a potential energy field having the periodicity of the lattice. As a result of these constraints, only certain energy states, or bands, are allowed for the charge carriers. In addition, the relationship between the energy and the velocity of the carriers is not the simple $1/2\ mv^2$ of the free particle, but is more complex. An approximation which works satisfactorily in conventional semiconductors is to characterize the charge carriers by a" effective mass m^*.[24b]

Taking into account mcst of the complexities we have mentioned, one still obtains a" expression equating the Hall coefficient to the inverse of the carrier concentration for the case where conduction occurs by means of carriers of a single type, i.e. all

electrons or all holes of a single effective mass. This means that conduction is in a single band. The more complex relationships for multiband conduction will be discussed later.

6. Single-Band Conduction

The expression for the Hall coefficient under the above conditions is commonly written in the form

$$R = r/ne \qquad \text{(practical units, } e < 0 \text{ for electrons).} \tag{46}$$

The Hall coefficient factor, r, depends on the nature of the scattering, the hand structure, the statistics characterizing the distribution of velocities of the carriers, and the magnetic field strength. Fortunately it depends rather weakly on these factors, and its value is usually within, say, 704 of unity.

To develop some relationships involving the Hall coefficient, we consider an isothermal and isotropic solid with $\vec{H}$ along the z axis and J along the x axis. The general transport relationships, given previously as Eq.(7), then reduce to

$$J_x = \sigma_{xx}E_x + \sigma_{xy}E_y, \qquad \begin{cases} H \equiv H_z \\ J \equiv J_x \end{cases} \tag{47}$$

$$0 = -\sigma_{xy}E_x + \sigma_{xx}E_y. \qquad \sigma_{ij} \equiv \sigma_{ij}(H) \tag{48}$$

The above relations plus the definitions of the Hall coefficient R and the conductivity σ allow us to express these quantities in terms of the components of the conductivity tensor, namely

$$R = E_y/J_xH = \rho_{xy}/H = \sigma_{xy}/(H[\sigma_x^2 + \sigma_{xy}^2]), \tag{49}$$

$$\sigma = J_x/E_x = 1/\rho_{xx} = [\sigma_{xx}^2 + \sigma_{xy}^2]/\sigma_{xx}. \tag{50}$$

It is to be noted that for the practical system of units, H (in gauss) is to be divided by 10^8 . From Eqs.(48)-(50). in conjunction with Eq.(5), some very useful relations can be derived:

$$E_y/E_x = \sigma_{xy}/\sigma_{xx} = R\sigma H = \tan 9. \tag{51}$$

An important attribute of charge carriers in a solid is their mobility, defined in the simple case as their drift velocity per unit electric field. The usual mobility, or more precisely the conductivity mobility, is commonly designated by μ and is related to the conductivity σ as indicated below:

$$\sigma = ne\mu. \tag{52}$$

It is also customary to define a Hall mobility μ^H by the relation

$$\mu^H = R\sigma . \tag{53}$$

From this definition plus Eq.(51), we see that

$$R\sigma = E_y/HE_x = (\tan\theta)/H, \tag{54}$$

$$\mu^H H = \tan\theta. \tag{55}$$

Thus we see that the Hall mobility bears a simple relation to the Hall angle. By use of Eq.(46) with (52) and (53), we establish the interesting finding that the Hall coefficient factor is the ratio of the Hall and conductivity mobilities, thus

$$r = \mu^H/\mu . \tag{56}$$

For many common semiconductors. the scattering of the charge carriers can be characterized by a relaxation time τ, which in most instances is energy-dependent. In such cases the conductivity mobility can be expressed by[25]

$$\mu = (e/m^*)\langle\tau\rangle , \tag{56a}$$

where $\langle\tau\rangle$ is the average of the relaxation time over the distribution of the charge carriers in energy. The Hall coefficient factor can also be related to averages of τ. For the case of weak magnetic fields, the result assumes the simple form[25]

$$r = \langle\tau^2\rangle/\langle\tau\rangle^2, \quad \mu^H H/10^8 \ll 1 \tag{57}$$

Both Eqs.(56a) and (57) assume a constant effective mass, that is, the existence of spherical energy surfaces and parabolic bands. Spherical energy surfaces ensure an isotropic effect mass; parabolic bands (the case where the charge-carrier energy is a quadratic function of momentum or wave vector) ensure that the effective mass does not vary with energy. If the preceding conditions are not met, then the m* in Eq.(56a) is also subject to the averaging. The case of nonspherical energy surfaces will be discussed shortly. The theory for nonparabolic bands has been presented by KoIodziejczak, [25a] with detailed treatment of the case of spherical energy surfaces and nonparabolic bands-representative of the conduction band in InSb.

Going back now to Eq.(57), we see that the value of r is affected by the nature of the scattering process. It is also magnetic-field dependent, approaching unity in the strong field region. Because of this behavior, it should be possible in principle to

determine r from the ratio of the Hall coefficients measured at the given field and at the strong-magnetic-field limit. In practice, however, the strong field must not be so great that quantum effects become significant.[26] This means that we require that $\hbar eH \ll kT$. Here $\hbar$ is the Dirac-Planck's constant, k is Boltzmann's constant, and T is the absolute temperature. Also the perfection of the measurement sample must be extremely good, since even minor inhomogeneities can cause aggravated problems at high magnetic field strengths.

A summary of how some common scattering mechanisms affect the Hall coefficient factor[26a] is shown in Table I. All entries except the last are for classical statistics, i.e. the Fermi level ζ lies outside the conduction or valence bands. The table also includes results for an arbitrary variation of the relaxation time τ with energy, given by the parameter λ. The value $\lambda = -1/2$ applies to acoustic phonon scattering, and $\lambda = 3/2$ is for scattering by ionized impurities. If the semiconductor is highly degenerate, i.e. the Fermi level lies deep within the band, then of course Fermi-Dirac statistics must be used. In the limit of high degeneracy it is essentially only those charge carriers of energy equal to the Fermi energy that contribute to conduction. Thus we have, in essence, a constant-τ case for all simple scattering mechanisms, and r is unity. In the strong field limit, the Hall coefficient factor is unity for all cases, provided that the magnetic field is not so huge that the separation between the Landau levels becomes comparable with kT. If the latter occurs, then quantum effects must be considered. Although the table shows only values of r in the limiting cases of zero and strong magnetic field strengths, the behavior of r can be calculated for arbitrary values of μ_H^H. However, the results are considerably more complicated.[26b]

As we implied in our discussion following Eq.(57), nonsphericity in the energy surfaces and nonparabolic dependences of charge-carrier energy on wave vector can also affect the Hall coefficient factor. We shall not take time to discuss the latter situation, a reference thereto having already been cited.[25a] To illustrate the effect of nonsphericity in the energy surfaces of a semiconductor, we include an anisotropy factor in the previous expression for r. Thus

$$r(H) = f(\tau,H)a(H). \qquad (58)$$

Here the factor $f(\tau,H)$ represents the contribution of the scattering processes to r(H), and the factor $a(H)$ is the result of nonisotropic energy surfaces. At very weak magnetic fields, $f(\tau,H)$ becomes the ratio of relaxation time averages shown before in Eq.(57), and the anisotropy factor approaches a value which we call a_o. At strong fields, $a(H)$ approaches unity.

Table I. Effect of Scattering on r
(Spherical Energy Surfaces)

Type of Scattering	Degree of Degeneracy	$(\mu_H^H/10^8<<1)$	$(\mu_H^H/10^8>>1)^*$
Acoustic phonon	None (ζ/kT <<0)	1.18 (i.e., 3 π/8)	1
Ionized impurity	Ditto	1.93 (i.e.. 315 π/512)	1
Constant-T	"	1	1
$\tau \sim \varepsilon^\lambda$	"	$[3\sqrt{\pi}/4][(2\lambda+3/2)!][\lambda+3/2)!]^{-2}$	1
Arbitrary	Highly degenerate ($\zeta/kT >> 1$)	1	1

*But must have $\hbar eH/10^8 m^* << kT$ so that quantum effects are negligible.

$$r_o = \langle r^2\rangle/\langle r\rangle^2 a_o, \quad \text{for } \mu^H H/10^8 \ll 1, \tag{58a}$$

$$r_\infty = 1, \quad \text{for } \mu^H H/10^8 \gg 1. \tag{58b}$$

For an energy surface represented by a system of ellipsoids arranged to possess cubic symmetry, a_o can be expressed in terms of the principal effective masses characterizing the ellipsoids, as shown in Eq. (59).[27]

$$a_o = 3(1/m_1^* m_2^* + 1/m_2^* m_3^* + 1/m_3^* m_1^*)/(1/m_1^* + 1/m_2^* + 1/m_3^*)^2 \tag{59}$$

For ellipsoids of revolution (spheroids), such as are characteristic of the conduction bands of germanium and silicon, Eq. (59) reduces to (61), where K is the ratio of longitudinal to transverse mass longitudinal being that along the axis of revolution of the ellipsoid.

$$\text{For spheroids: } ml^* = m_2^* = m_\perp^*, \quad m_3^* = m_\parallel^* \tag{60}$$

$$a_o = \frac{3K(K+2)}{(2K+1)^2}, \quad K \equiv \frac{m_\parallel^*}{m_\perp^*} \tag{61}$$

A plot of a_o as a function of K is shown in Figure 3. We note that for prolate spheroids, representative of common semiconductors such as germanium and silicon, a_o lies between 0.78 and 1. For pie-shaped surfaces, however, it can get quite low. Numerical values of a_o for various bands in germanium and silicon are listed in Table II. The results for the warped heavy-mass bands were obtained from the data of Lax and co-workers.[28] For the light-mass valence bands, which are approximately spherical, a_o is essentially unity.

The reader will note that thus far I have discussed the Hall anisotropy factor only in the limit of very weak and very strong magnetic fields. At intermediate field strengths, things can be very complicated. For example, in the spheroid model, results can depend on what particular arrangement of spheroids is used to provide the cubic symmetry, i.e. (a) 3 or 6 spheroids with major axes along the [100] directions, (b) 12 spheroids along [110] directions, or (c) 4 to 8 spheroids along [111] directions.*[9] Furthermore, as might be expected, moving the magnetic field vector off a direction of high symmetry can cause significantly different variations in the magnetic field dependence of the Hall coefficient. As an example of this, we present in Figure 4 data taken by Mason and co-workers on 3.18 ohm-cm germanium.[30] It is seen that moving H off the [001] direction can exert a strong influence on the magnetic-field dependence of R.

In the case of warped energy surfaces, the magnetic-field dependence of r can possess maxima when certain admixtures of

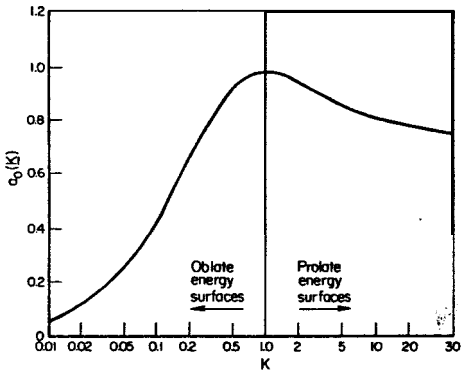


Fig. 3. Dependence of anisotropy factor in μ^H/μ on the anisotropy of the effective mass for a family of spheroidal energy surfaces arranged so as to possess cubic symmetry. The quantity K is the ratio $m_\parallel^*/m_\perp^*$ (after Herring).[27]

Table II. a_o in Germanium and Silicon

Conduction band:

$a_o = 3K(K+2)/(2K+1)^2$

Ge: $K \simeq 1$, $a_o \simeq 0.79$

Si: $K \simeq 4.7$, $a_o = 0.87$

Warped heavy mass valence band:

Ge: $a_o \simeq 0.71$

Si: $a_o \simeq 0.76$

Light mass valence band:

$a_o = 1.$

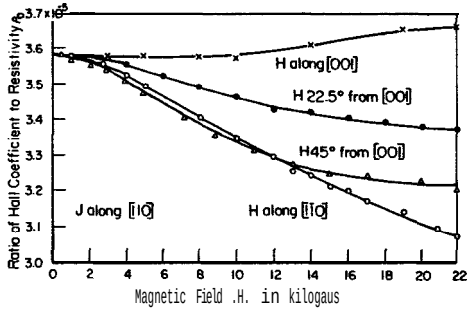


Fig. 4. Variation of $R(H)/\rho_o$ with H for four directions of the magnetic field in 3.18 ohm-cm germanium (after Mason, Hewitt, and Wick).[30]

scattering mechanisms exist,[31] as is seen in Figure 5. Here β represents the degree of admixed ionized impurity scattering, and μ_L is the mobility representative of lattice scattering in a spherical band. We see from the figure that the warping produces a significant lowering of the Hall coefficient factor r at weak magnetic fields. This is apparent in the curves for small β, indicative of the case where acoustic phonon scattering is strongly predominant. The admixture of increasing amounts of ionized impurity scattering introduces conspicuous changes, eventually producing striking maxima. With regard to Figure 5, I would point out that the theoretical development on which the curves shown there are based involves approximations which could stand improvement. Nevertheless, the maxima are real, as is borne out by experimental data to be shown later. I present the results to illustrate the sensitivity of the Hall effect to complexities in scattering and warping in band structure.

A general discussion of the effect of warped energy surfaces on μ_H/μ has been given by Shockley,[32] who presents a helpful physical explanation for the lowering of μ_H/μ. He points out how, for electrons in certain states in solids with re-entrant energy surfaces, the ratio μ_H/μ can become negative. Anisotropy factors have been calculated by several investigators for a variety of planar-faced energy surfaces.[33-35] Values of a_o are 1/2 for a cube and 2/3 for an octahedron,[34,35] with approaches toward unity as the number of faces is increased.[36] Effects of the rounding of edges and corners have also been considered.[37]

It is to be noted that for simplicity in presentation I have chosen to deal separately with the influence on the Hall coefficient

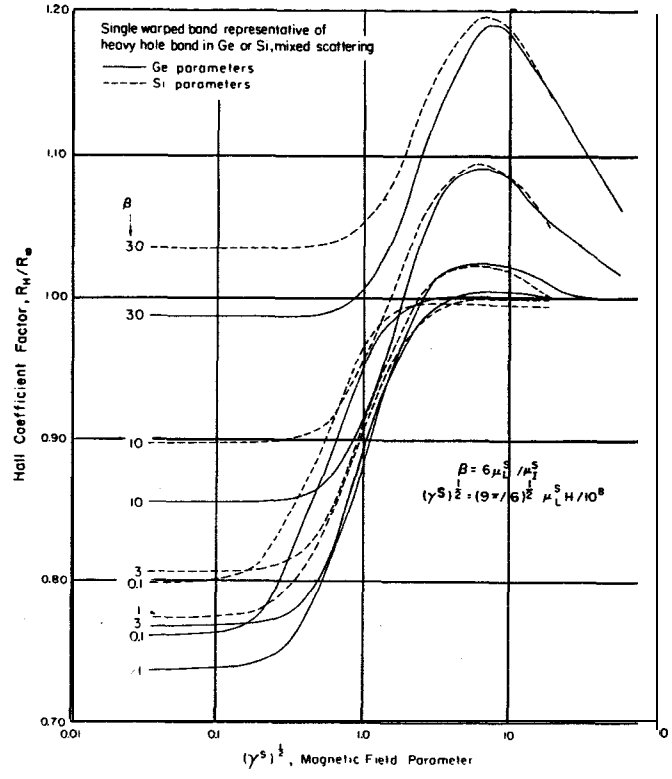


Fig. 5. Hall coefficient factor for a single warped band. The solid lines are for parameters representative of the heavy mass band in germanium, as determined from cyclotron resonance studies. The dashed lines are applied to silicon. The parameter β measures the degree of ionized impurity scattering (after Beer and Willardson).31

factor of such entities as nature of the charge-carrier scattering energy band anisotropy, and multiband effects (the latter to be

discussed in Section 7). However, the same basic consideration underlies all of these cases-namely a situation where certain charge carriers possess different Hall angles. For this reason, Allgaier has referred to r as a mixing factor,[36] since it is the mixing of charge carriers with different Hall angles which causes r to differ from unity.

In concluding this section, I want to emphasize that careful studies of Hall effect can tell us something about the predominant scattering mechanisms in the semiconductor, and also they can supply information about any anistropies in band structure. In obtaining information about scattering, one needs especially to pay attention to variations in Hall coefficient factor with temperature and with magnetic field; for studies on band structure, the variations with magnetic field and with sample orientation are significant.

7. Multiband Conduction

Let us now add another complication-the case where conduction occurs in two or more bands. Then every component of the conductivity tensor receives contributions from each of the bands, and these must be summed. To simplify the equations, we consider the case of an isotropic solid (or cubic crystal with the coordinate axes along the cube directions) and-place the current in the x direction and the magnetic field in the z direction. Then the application of the simplified transport equations [(47) and (48)] shown before, yields

$$J_x = [\sum_i \sigma_{xx}^{(i)}(H)]\, E_x + [\Sigma \sigma_{xy}^{(i)}(H)]\, E_y, \qquad H \equiv H_z,\ J \equiv J_x \qquad (62)$$

$$0 = [\sum_i -\sigma_{xy}^{(i)}(H)]E_x + [\Sigma \sigma_{xx}^{(i)}(H)]E_y,$$

where $\sigma_{ik}^{(i)}$ refers to band i.

For the summations over the various bands contributing to the conduction process, σ_{xx} is always positive and, for non-pathological bands (the case of certain re-entrant energy surfaces has been discussed by Shockey),[32] σ_{xy} is negative for electrons and positive for holes. We consider again only the limiting cases of the Hall coefficient for very weak and very strong magnetic fields. For electrons and holes of concentrations n and p respectively, the results are[38]

$$R_o = -\frac{r_{on}\mu_n^2 n - r_o \mu_p^2 p}{|e|(\mu_n n+\mu_p p)^2},\quad R_\infty = -\frac{1}{|e|(n-p)}, n\neq p. \qquad \begin{matrix} n,\text{electrons} \\ p,\text{holes} \end{matrix} \qquad (63,64)$$

In the case of light and heavy holes, as is representative of the valence bands of a large number of semiconductors, the expressions for the Hall coefficient are

$$R_o = \frac{r_{o1}\mu_1^2 p_1 + r_{o2}\mu_2^2 p_2}{|e|(\mu_1 p_1 + \mu_2 p_2)^2},\quad R = \frac{1}{|e|(p_1+p_2)}. \qquad \begin{matrix} 1,\text{heavy holes} \\ 2,\text{light holes} \end{matrix} \qquad (65,66)$$

The interesting observation, seen in both Eqs.(63) and (65). is that in the expression for the weak-field Hall coefficient the carrier densities are weighted by the squares of the respective mobilities. Thus a small concentration of minority carriers can exert a large effect if their mobilities are large. As the magnetic field is increased, however, the weighting factors become less controlling (this can be seen if the above sets of equations are developed as a function of magnetic field[39] rather than in the simplified forms representative of merely the low- and the high-field limits), and the mobility weighting factors are absent altogether in the high-field limit [as is seen in Eqs.(64) and (66)].

Concentrating on Eq.(63) for a moment, we see that a strongly p-type semiconductor with p=6400n will actually have a negative Hall coefficient R_o if $\mu > 80\mu_h$. As the magnetic field is increased, the Hall coefficient wilf cross over to positive values as approximation (64) begins to apply. An excellent example of this behavior[39a] is provided by the data on InSb shown in Figure 6. Because of the relatively high electron mobility, the specimen must be cooled to quite a low temperature before the extrinsic p-type Hall coefficient is attained. Actually, InSb has both heavy and light holes, but the electron mobility exceeds that of even the light holes. In semiconductors of this type, the Hall coefficient crossover moves to higher temperatures (i.e. larger values of electron density) as the magnetic field is increased. This is the behavior we mentioned above in connection with the transition from Eq.(63) to (64). and it is caused by the fact that the high-mobility carriers (which are closer to the strong-field approximation region) are electrons. Therefore an increase in magnetic field requires a greater relative density of electrons (hence a higher temperature) to maintain the Hall coefficient null.

An opposite situation to that discussed above is found in p-type geramnium[39b] (see Fig. 7). Here it is the mobility of the light holes which is greater, by a significant amount, than that of the electrons. The Hall coefficient crossover therefore moves to lower temperatures (i.e. a larger relative density of light holes as compared to electrons) as the magnetic field is increased. This behavior is quite noticeable in the data presented in Figure 7. The dashed lines shown in the lower part of the figure represent a calculation for the case where there are no light holes, and therefore

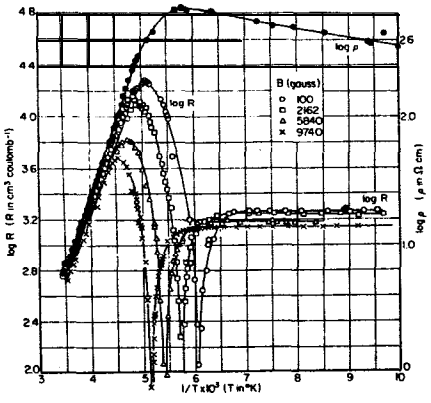


Fig. 6. Temperature dependence of the Hall coefficient of a p-type InSb speciment ($p \simeq 4 \times 10^{15}$ cm-3) at different magnetic field strengths. The resistivity at zero field is also shown (after Howarth. Jones, and Putley).[39a]

the electron mobility is the larger (since it significantly exceeds that of the heavy holes). This was the situation in the InSb example. We also note in the figure that when the magnetic field becomes so large that all carriers approach the strong-field region, both curves assume similar slopes. Calling attention again to Fig. 6, I wish to point out that Hall data of the type shown can provide "a with simple relationships involving carrier densities and electron-hole mobility ratios. Of value, besides the crossover temperature, is the ratio of the Hall maximum to the value at the extrinsic plateau. For a simple model with only electrons and one type of hole and a single acceptor level of density NA and negligible activation energy, the carrier concentrations can be expressed by

$$p = n + N_{A'} \quad \text{or} \quad \frac{p}{NA} = \frac{n}{N_A} + 1 = X + 1, \quad \text{where } X \equiv n/N_A \tag{67}$$

$$n = XN_{A'} \qquad p = (X + 1)N_A. \tag{68}$$

For simplicity, we assume that the Hall coefficient factors are the same for each band so that, with the "se of relations (68), Eq.(63) can be put in the form

$$R_o = -\frac{r_o}{|e|N_A}\frac{(c^2-1)X-1}{[(c+1)X+1]^2}, \quad c \equiv \mu_n/\mu_p, \quad r_{on}=r_{op}\equiv r_0, \tag{69}$$

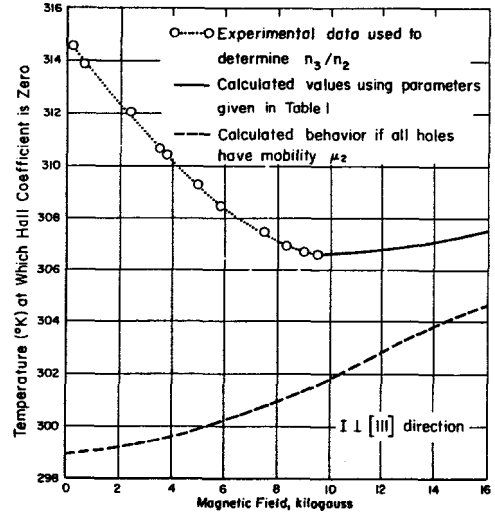


Fig. 7. Experimental data and calculated results of Hall coefficier crossover temperatures in p-type germanium ($p \simeq 4 \times 10^{13}$ cm^{-3}) as a function of magnetic field (after Willardson, Harman. and Beer).[39b]

It follow at once that at the Hall coefficient null

$$X = 1/(c^2-1), \text{ i.e. } n^{null} = N_A/(c^2-1). \tag{70}$$

With use of elementary calculus, we find that at the Hall coefficient maximum

$$x = 1/(c-1), \quad \text{i.e. } n^{max} = N_A/(c-1). \tag{71}$$

One additional expression-simple, yet very useful-follows from the above equations. This is the ratio between the values of the weak-field Hall coefficient at its maximum and at its extrinsic plateau, namely

$$R_o^{max}/R_o^{ext} = -(c-1)^2/4c. \tag{72}$$

This relationship depends only on the mobility ratio of the electrons and holes.

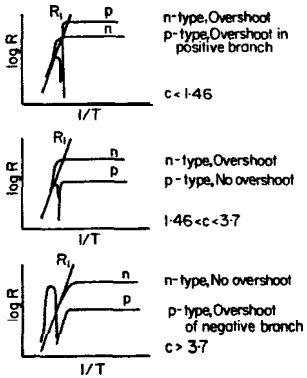


Fig. 8. Hall effect in mixed electron-hole conduction showing effect of different values of the electron-hole mobility ratio, **c** (after Putley).40

In concluding the discussion on p-type material, I cannot emphasize too strongly the importance-particularly for studies done on unfamiliar materials-of measuring magnetic-field dependences and temperature variations of the Hall effect. Think how ridiculous a result could ensue if one applied the simple formula R = **1/pe** when data were taken at magnetic fields which placed the material in the vicinity of the crossovers shown in Fig. 6!

The mobility ratio also influences the nature of the temperature dependence of the Hall coefficient in n-type material in the transition region between the extrinsic plateau and the intrinsic slope. This behavior has been studied by Putley.40 and his results are shown in Figure 8. We see that the effects of the different mobility ratios are quite noticeable.

Although the crossovers are not involved, conspicuous two-band effects can be observed in extrinsic p-type semiconductors when light and heavy hole bands exist, as is the case, for example, in germanium, silicon, and comm nIII-V compound semiconductors. Figure 9 shows the excursion of the Hall coefficient in extrinsic p-type germanium as the magnetic field is varied from the weak-field to the strong-field regions[39b]--Eq.(65) to Eg. (66). Note that for measurements done at constant magnetic fields of several kilogaus or less, the Hall coefficient shows a large temperature variation which does not exist if the data are taken at around ten kilogauss. This apparently anomalous temperature variation is more

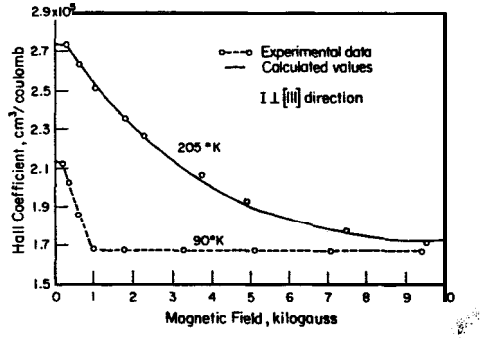


Fig. 9. Hall coefficient as a function of magnetic field at two temperatures for a p-type germanium specimen (after Willardson, Harman, and Beer).

explicitly revealed[40a] in the data plotted as a function of temperature in Figure 10. It is important to realize that in this case the naive application of the simple Hall effect relation R = **1/pe** would lead one to think that the hole density decreased substantially with increasing temperature-a thoroughly puzzling state of affairs. Actually what happens is that the temperature increase moves the material more toward the weak-field region so that the effect of the light holes becomes more controlling. But, you may say, the magnetic field was kept constant. The point to be emphasized is that it is the product **μH** which determines the existence of the weak- or strong-field region. Hence you may religiously maintain H constant, but when the mobility changes with temperature, as it does in most semiconductors, a temperature excursion will include the effects of a magnetic field variation. In contrast to the conduction resulting from the light- and heavy-hole bands, the transport due to the electrons-where only a single band is involved-shows normal behavior for **Rσ/μ,** according to the lower curve of Figure 10. Thus once again we see compelling reasons for including magnetic field dependences in any measurement of Hall effect.

Two-band effects can also be important in a number of extrinsic semiconductors where the bands in question are separated in energy but where carriers are activated into the higher-lying band at the measurement temperature. Then the carrier-concentration terms in **Eqs.(65)** and (66) are temperature-dependent. Common examples are found in a number of compound semiconductors, such as

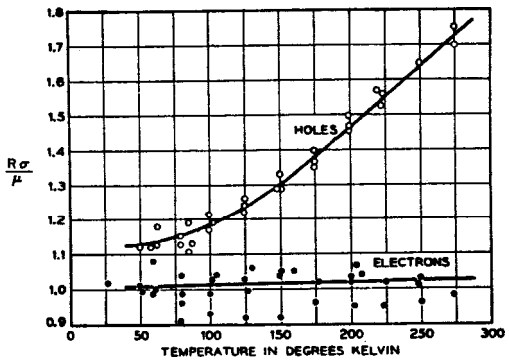


Fig. 10. Temperature dependence of $R\sigma/\mu$ for different specimens of n- and p-type germanium (after Morin).[40a]

the conduction bands in GaSb and, at high temperatures, GaAs. Note that in the III-V materials, this effect is in addition to the influence of the light- and heavy-h& bands, which are degenerate at the center of the Brillouin zone, and therefore there is no activation energy involving the light and heavy holes. Detailed analyses of Hall effect as a function of temperature in materials having conduction bands such as those mentioned above, have provided information on the separation in energy of the bands in question and on the ratio of the mobilities in those bands. Because of the separation in energy of the bands in question, it is readily possible that the charge carriers in one band may be degenerate (i.e. require the use of Fermi-Dirac statistics), while those in the other band can be described by the classical statistics.

8. Two-Band Conduction with Warping and Mixed Scattering

Lest even at this point, one or two of the more blase of my readers might become complacent, let me show some weird effects that can occur in the magnetic-field dependence of the Hall coefficient when we have two-band conduction involving a light-hole band and a warped heavy-hole band, each with different admixtures of acoustic phonon and ionized impurity scattering. Time and space will not allow much discussion-just a look at some graphs. Figure 11 shows p-type germanium with an interesting minimum. But let us not be partial to minima. We can take band characteristics representative of silicon and get some pretty maxima, as seen in

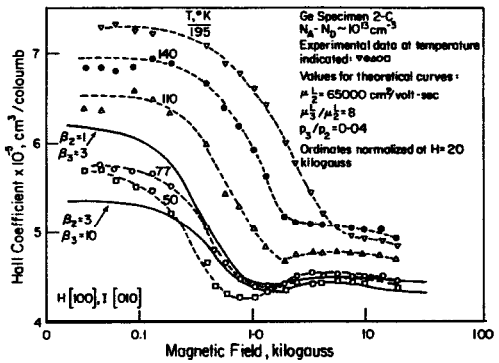


Fig. 11. Magnetic field dependence of Hall coefficient in p-type germanium for various temperatures Theoretical curves (solid lines) are based on μ_2 = 65,000 cm^2/volt-sec, representative of the mobility at 77°K and on normalization of ordinates at 20 kgauss for the $\beta_2 = 1$ curve to fit the experimental curves, the curve of $\beta_3 = 3$ being slightly depressed for clarity. Subscripts 2 and 3 refer to heavy and light-mass bands respectively (after Beer).[41c]

Figure 12. .A similar type of behavior was found to occur in type IIB diamond and in AlSb. Figure 13 presents such data. The resemblance to silicon is quite striking.

9. Inhomogeneities in the Measurement Sample

Finally, I want to devote a little time to show how inhomogeneities in the measurement sample can produce some disturbing results. However, there is only time to state that certain inhomogeneities can produce an admixture of Hall and magnetoresistivity effects-thereby causing large magnetic-field dependences for the Hall coefficient. This behavior is illustrated by a step function model-the measurement parallelepiped having a discontinuity in Hall coefficient and resistivity. For $x < 0$, the respective values are denoted by Rl and pl; for $x > 0$, they are R_2 and ρ_2. These values are those of the material in the applied magnetic field, although in many cases R may vary only slightly with field or be constant. We shall omit details of derivation and merely present the final expression for the Hall coefficient as measured across potential orobes placed at x = 0, which defines

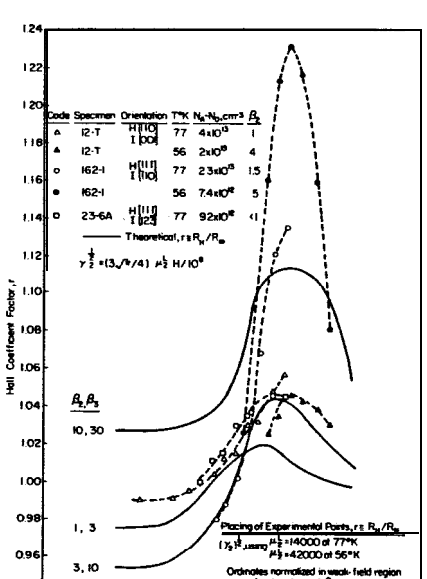


Fig. 12. Experimental data (dashed lines) indicating magnetic-field variation of Hall coefficient factor in extrinsic p-type silicon for two temperatures. Theoretical predictions for different amounts of ionized impurity scattering are shown by solid lines (after Beer).[41c]

the plane of discontintuity in R and ρ. The result is[42]

$$R_{x=0}^{meas} = (R_1\rho_2 + R_2\rho_1)/(\rho_1 + \rho_2). \qquad (73)$$

Inasmuch as ρ can have a large variation with H, we see that this admixture of Hall and magnetoresistive effects may produce a large magnetic-field variation in the measured value of R, even if R_1 and R_2 should be essentially independent of H. The magnitude of the measured Hall coefficient may increase or decrease with magnetic field. For it to increase, the following condition must be

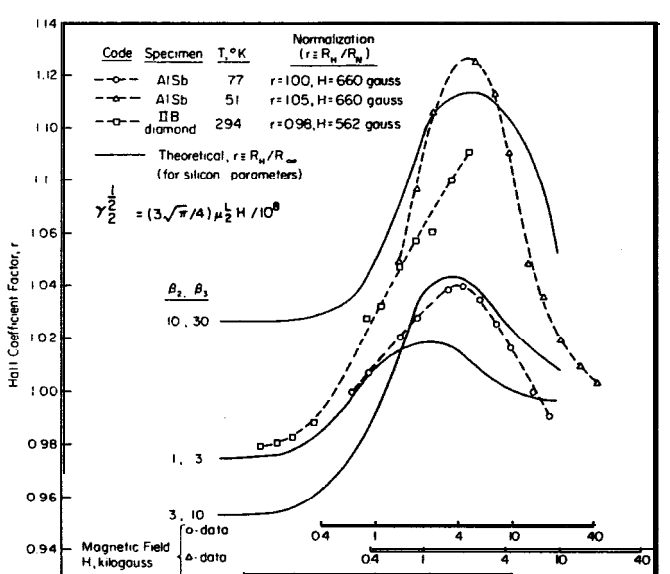


Fig. 13. Experimental data (dashed lines) shoving magnetic-field dependence of Hall coefficient factor for p-type diamond and AlSb. Superimposed are solid-line theoretical curves (cf. Fig. 12) representative of silicon (after Beer).[41c]

satisfied:[42]

$$\frac{1}{R}\frac{\partial R}{\partial H} = \frac{\rho_1\rho_2}{\rho_1 + \rho_2}\frac{R_2 - R_1}{R_2\rho_1 + R_1\rho_2}\left[\frac{1}{\rho_1}\frac{\partial \rho_1}{\partial H} - \frac{1}{\rho_2}\frac{\partial \rho_2}{\partial H}\right]. \qquad (74)$$

where $R = R_{x=0}^{meas}$ of Eq.(73). In deriving the above relationship, we have neglected any changes in Hall coefficient with magnetic field in each material, since these changes are usually quite small compared to the resistive changes. Equation (74) tells us that the measured Hall coefficient will increase with magnetic field if the inherent magnetoresistanc$(1/\rho)\partial\rho/\partial H$, is larger in the region

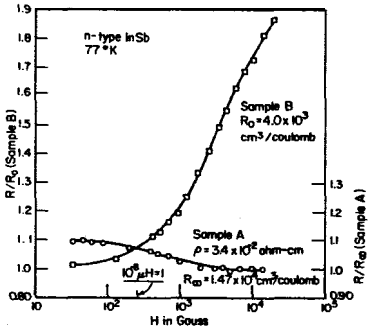


Fig. 14. Magnetic-field dependence of the normalized measured Hall coefficient in two high-purity n-type InSb samples. Sample A is homogeneous, but sample B exhibits a change in electron density of a factor of 10 along its length. Carrier concentrations are: sample A: 4.3×10^{14} cm^{-3}; sample B, 5×10^{14} to 5×10^{15} cm^{-3} (after Bate, Bell, and Beer).[42]

having the smaller inherent Hall coefficient.

An illustration of the type of behavior predicted by Eq.(73) is provided by two measurement samples of InSb—one of which was quite uniform, while the other exhibited a variation in carrier density ranging from $5x10^{14}$ to $5x10^{15}$ cm^{-3}. The data are shown in Figure 14. It is strinkingly evident that the presence of inhomogeneities can completely mask the normal variation in Hall coefficient factor with magnetic field.

Again we stress that much caution is necessary in interpreting Hall data taken on specimens of unknown character. One can obtain large distortions in apparent Hall mobility—either on the low or high side—depending on the nature of the inhomogeneities, placement of Hall and current probes, etc. An example of lowered mobility is seen in Figure 15 for the step-function resistivity variation. Mobility enhancement was studied by Wolfe and Stillman,[43,43a] who showed that multiple metallic inclusions can cause such behavior. The high apparent mobility is most commonly observed in compound semiconductors which are grown under metal-rich conditions. Temperature dependences of mobility as determined from the $R\sigma$ product for anomalous and normal GaAs samples were measured by Wolfe and co-workers, and are shown in Figure 16. Note that the ordinate is a logarithmic scale, so that the mobility augmentation is fairly large. Imagine the disappointment of a materials-preparation researcher who might first be led to believe that he had produced

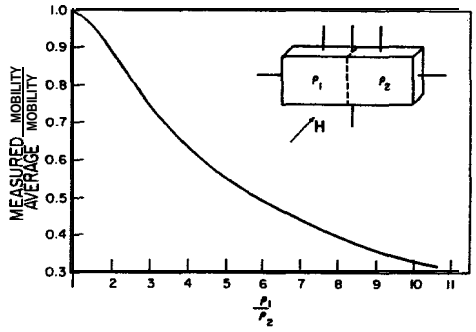


Fig. 15. Apparent Hall mobility as a function of the ratio of resistivities in a step-function model. Hall probes are placed on the plane of discontinuity. The normalization factor is the average of μ_1^H and μ_2^H (after Bate, Bell, and Beer).[42]

some super-duper once-in-a-lifetime material! I think it is abundantly clear that meaningful analyses of Hall data require checks on the homogeneity of the sample. Methods for doing this were mentioned earlier (Section 4).

10. Materials and Situations Requiring Special Treatment

Although we have covered a rather large number of complex situations that can be encountered in using Hall effect in a broad spectrum of practical cases, there are numerous types of materials and situations which require special treatment that has not been considered here. For example, certain metals have Fermi surfaces reminiscent of some kinds of modern sculpture. If open orbits are encountered, the behavior of the Hall coefficient with magnetic field and the interpretation of the Hall data are vastly complicated.[44] In ferromagnetic materials, an additional phenomenon comes into being—namely the "extraordinary Hall effect"—which is related to the magnetization.[45,46] At very strong magnetic fields in semiconductors, quantum effects appear[47] when the Landau level spacing becomes important—a situation which appears when $\hbar eH/m^*$ becomes larger than kT. Another consideration is hot carrier effects, which assume importance at strong electric fields when the current may no longer vary linearly with field.[48]

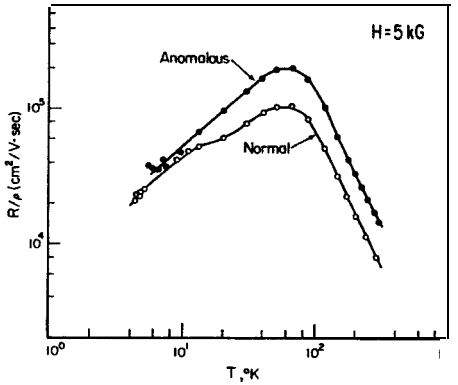


Fig. 16. Apparent mobility enhancement in GaAs. The anomalous specimen is believed to contain metallic inclusions, presumably gallium precipitates (after Wolfe et al.).[43a]

Finally, I would point out that the transport theory we have presented here is based on concepts which, in the simplest interpretation, may be related to the concept of a charge-carrier relaxation time or mean free path—quantities which are proportional to mobility.[49] There are materials, however, with very low mobilities. For such materials, I think it is apparent that our conventional transport theory will be in trouble when the mean free path becomes less than the interatomic distance.[50] A related consideration deals with the fact that conventional transport is treated by classical theory, based on the Boltzmann equation. Its validity requires that $\hbar/\tau$ be significantly greater than the Fermi energy.[49] For materials where some of the above conditions are not fulfilled, transport theory has been developed which is based on a hopping process.[50] Types of amorphous semiconductors are in the category requiring special treatment.

Some of the materials and situations mentioned above will be discussed later by other speakers at this conference.

V. CLOSING REMARKS

In closing, I would remark that we have seen how the Hall effect has far deeper substance and more arcane subtleties than one might ever guess from the simple free-electron relationships based on the Lorentz force. We have seen complexities and

unexpected confrontations in measurement, in theory, and in interpretation; perturbation of sought-after fields by contacts and specimen geometries, by ubiquitous thermal effects, and by higher order galvanomagnetic coefficients—along with the unique influence of material anisotropies, mixed scattering processes, light and heavy charge carriers, warped bands, and specimen imperfections. We have seen how all these matters conspire to unfold a drama and portrait revealing ever-changing facets and challenges. Yet what a wealth of information can be extracted through diligent effort and study. But after all, is this not an example of the kaleidoscope of beauty and color that springs forth in science?

If now we had glasses of champagne, we would drink a toast to Edwin Hall. We would do it in the words of Shakespeare "How infini in faculties" and in the poetry of Keats "A thing of beauty is a joy for ever."

ACKNOWLEDGMENT

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THE HALL EFFECT IN HEAVILY DOPED SEMICONDUCTORS

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ABSTRACT

The properties of the Hall effect in relatively pure, light doped semiconductors are now generally well understood. However the Hall effect and mobility in heavily-doped semiconductors, which have metal-like behavior, have been studied much less experimentally and are still not fully understood theoretically. This paper reviews the Hall effect and related properties of heavily-doped semiconductors for the cases of both uniform and non-uniform doping. The unexplained experimental results remain a challenge to theory but the empirical knowledge of the Hall effect in heavily-doped semiconductors is useful in solid-state device technology.

1. INTRODUCTION

The Hall effect, along with the electrical resistivity, have been and continue to be the key parameters used in investigations of the basic electrical conduction processes in semiconductor materials. In this paper, we discuss the results of the Hall effect and resistivity studies of the electrical conduction and carrier scattering mechanisms in a variety of semiconductor materials uniformly doped with high concentrations of known impurities. Surprisingly, the electrical conduction processes are even today not well understood in heavily doped semiconductors. The application of the Hall effect to the analysis of impurity concentration and distribution in non-uniformly doped semiconductors has proven to be a key tool in the investigation of diffusion and ion implantation processes. The application of the Hall effect to non-uniformly doped semiconductors is discussed along with examples of results